Problem 8(a). Determine the number of distinct 4-bead n-color necklaces if necklaces are considered the same when one is a rotation or reflection of the other.

Proof. Let X be the set of all 4-bead colorings using n colors.

We use Burnside's Lemma for $G \curvearrowright X$ where both are finite:

$$\#(\text{distinct } (G \curvearrowright X)\text{-orbits}) = \frac{1}{|G|}\sum_{g\in G} |\text{Fix}(g)|,$$

Here $X := \left\{ \begin{smallmatrix} x_3 & x_0 \\ x_2 & x_1 \end{smallmatrix} \mid x_1, x_2, x_3, x_4 \in [n] \right\}$ and $G = D_4 = \{e, r, r^2, r^3, F, Fr, Fr^2, Fr^3\}$.

The group D_4 is isomorphic to the following subgroup of S_4:

$$D_4 = \{(1)(2)(3)(4), (1234), (13)(24), (1432), (24)(1)(3), (12)(34), (13)(2)(4), (14)(23)\}.$$

We calculate the size of $|\text{Fix}(g)|$ by counting the number of elements (tuples in $[n]^4$) of X fixed by g, whose general form is given in the second column. Observe.

g	$Fix(g)-$ elements	$\lvert\text{Fix}(g)\rvert$
$(1)(2)(3)(4)$	$\begin{smallmatrix} d & a \\ c & b \end{smallmatrix}$	n^4
(1234)	$\begin{smallmatrix} a & a \\ a & a \end{smallmatrix}$	n
$(13)(24)$	$\begin{smallmatrix} b & a \\ a & b \end{smallmatrix}$	n^2
(1432)	$\begin{smallmatrix} a & a \\ a & a \end{smallmatrix}$	n
$(24)(1)(3)$	$\begin{smallmatrix} b & a \\ c & b \end{smallmatrix}$	r^3
$(12)(34)$	$\begin{smallmatrix} b & a \\ b & a \end{smallmatrix}$	r^2
$(13)(2)(4)$	$\begin{smallmatrix} c & a \\ a & b \end{smallmatrix}$	r^3
$(14)(23)$	$\begin{smallmatrix} a & a \\ b & b \end{smallmatrix}$	r^2

Thus,

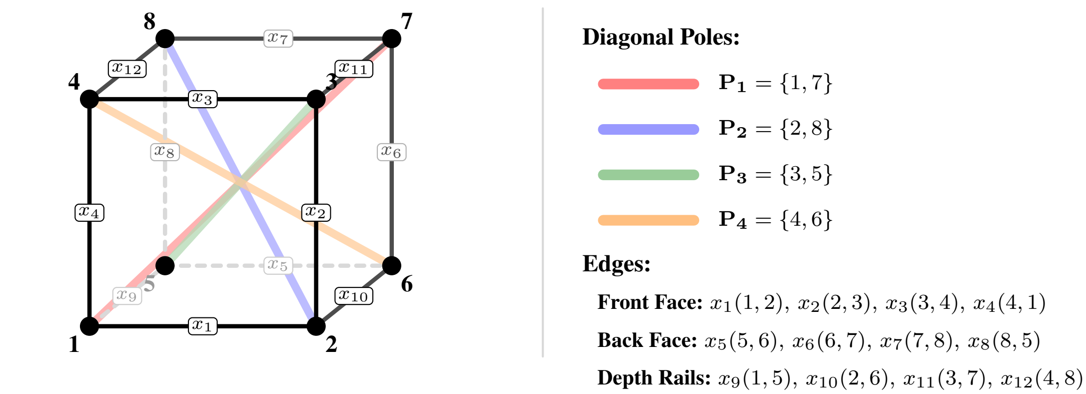
$$\#(\text{different n-color 4-necklaces up to } D_4) = \#(\text{distinct } (D_4 \curvearrowright X)\text{-orbits}) = \frac{1}{8}\left(n^4 + 2n^3 + 3n^2 + 2n\right).$$

$\square$

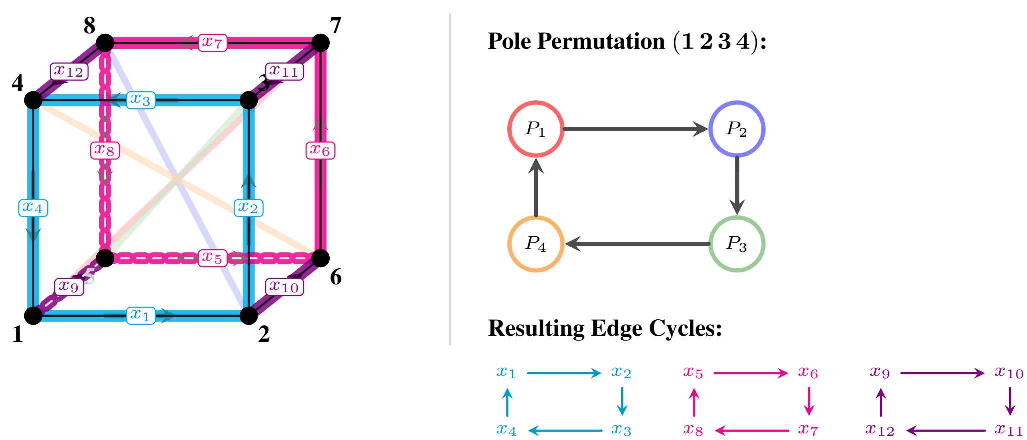
- **Sagan—Chapter 6:** Ex. 8(a), 9(a), 11 two parts, 15 two parts, 31(a,b)
- **Fulton—Chapter 7:** Ex. (2-3), calculate S_4 table

Problem 9(a). Find the number of distinct edge n-colorings of a cube under rotations using Burnside's Lemma.

Proof. Let X be the set of all edge n-colorings of the edges of a cube $X \longleftrightarrow [n]^{12}$ and G be the rotational symmetries of X. S_4 permutes the set of 4 diagonal poles inside a cube, and those permutations describe rotations of the cube. So these are the same permutations; $G \cong S_4$. The edge labels, vertices, and corresponding poles are shown below:



So we can just map the action of S_4 on the poles to permutations of vertices, which we can then map to permutations on edges X. We then obtain $c(g)$ via this isomorphism $s \in S_4 \leftrightarrow v \in V \leftrightarrow g \in G$. Visually this is very simple and we show an example below.



So $c(g) = 3$ as shown from the resulting edge cycles, and then $|\text{Fix}(g)| = n^(c(g)) = n^3$. We show the rest of the computations below.

Geometric Rotation Class	**Pole Permutation** $g \in S_4$	$c(g)$	$\lvert\mathrm{Fix}(g)\rvert$
1) Identity (1 element)			
Identity	$(1)(2)(3)(4)$	12	n^{12}
2) Face-Center Rotations (9 elements)			
90° around Z-axis	$(1\,2\,3\,4)$	3	n^3
270° around Z-axis	$(1\,4\,3\,2)$	3	n^3
180° around Z-axis	$(1\,3)(2\,4)$	6	n^6
90° around Y-axis	$(1\,4\,2\,3)$	3	n^3
270° around Y-axis	$(1\,3\,2\,4)$	3	n^3
180° around Y-axis	$(1\,2)(3\,4)$	6	n^6
90° around X-axis	$(1\,3\,4\,2)$	3	n^3
270° around X-axis	$(1\,2\,4\,3)$	3	n^3
180° around X-axis	$(1\,4)(2\,3)$	6	n^6
3) Edge-Center $180°$ **Flips (6 elements)**			
Flip via x_1/x_7 midpoint	$(1\,2)(3)(4)$	7	n^7
Flip via x_3/x_5 midpoint	$(3\,4)(1)(2)$	7	n^7
Flip via x_2/x_8 midpoint	$(1\,3)(2)(4)$	7	n^7
Flip via x_4/x_6 midpoint	$(2\,4)(1)(3)$	7	n^7
Flip via x_9/x_{11} midpoint	$(1\,4)(2)(3)$	7	n^7
Flip via x_{10}/x_{12} midpoint	$(2\,3)(1)(4)$	7	n^7
4) Vertex-Center $120°/240°$ **Spins (8 elements)**			
120° spin around Pole P_1	$(1)(2\,3\,4)$	4	n^4
240° spin around Pole P_1	$(1)(2\,4\,3)$	4	n^4
120° spin around Pole P_2	$(2)(1\,4\,3)$	4	n^4
240° spin around Pole P_2	$(2)(1\,3\,4)$	4	n^4
120° spin around Pole P_3	$(3)(1\,2\,4)$	4	n^4
240° spin around Pole P_3	$(3)(1\,4\,2)$	4	n^4
120° spin around Pole P_4	$(4)(1\,3\,2)$	4	n^4
240° spin around Pole P_4	$(4)(1\,2\,3)$	4	n^4

By Burnside's Lemma, we average the number of fixed configurations across the 24 pole permutations of S_4:

$$\#(\text{distinct edge } n\text{-colorings}) = \#(\text{distinct } (G \curvearrowright X)\text{-orbits}) = \frac{1}{|S_4|} \sum_{g \in S_4} |\mathrm{Fix}(g)|$$

Adding all n-polynomials in the last row of the table we get the following

$$\#(\text{distinct edge } n\text{-colorings}) = \frac{1}{24}\left(n^{12} + 3n^6 + 6n^7 + 8n^4 + 6n^3\right)$$

□

Problem 11(a). Compute the cycle index polynomial for $\langle(1\cdots n)\rangle \cong Z_n \curvearrowright X = [n]$.

Proof. Recall: For a finite group G acting on a finite set X with $|X| = n$, the cycle index polynomial is

$$Z(G \curvearrowright X) = \frac{1}{|G|}\sum_{g\in G}\prod_{i=1}^{n} a_i^{c_i(g)},$$

Let $\sigma = (1\,2\,\ldots\,n) \implies G = \langle\sigma\rangle \cong Z_n$ once more. We determine the cycle structure of σ^k permuting $[n]$ for each $k = 1, \ldots, n$.

Take $[n]$ to be $\mathbb{Z}_n$. Then, to compute the length of any cycle in the 'maximal length decomposition' of σ^k, fix $i \in [n]$ and apply σ^k repeatedly to see in how many steps it wraps back to itself; to see how long the cycle it is contained in is.

$$\sigma^k(i) = i + k \pmod n \implies (\sigma^k)^j(i) = i \text{ when } i + jk \equiv i \pmod n \iff jk \equiv 0 \pmod n.$$

So we see that for any i, the cycle length is the minimal number j such that $jk \equiv \pmod n$; $|k|_{\mathbb{Z}_n}$, which is not only well-known to be $|k|_{\mathbb{Z}_n} = \frac{n}{\gcd(k,n)}$ but it is also independent of which $i \in [n]$ being permuted. So σ^k must partition $[n]$ into $m = \frac{n}{\frac{n}{\gcd(k,n)}} = \gcd(n,k)$ equal parts of length $\frac{n}{\gcd(k,n)}$. In more precise terms, σ^k decomposes into exactly $\gcd(n,k)$ cycles which all have length $\frac{n}{\gcd(n,k)}$. That is,

$$\sigma^k \text{ contributes } a_{\frac{n}{\gcd(k,n)}}^{\gcd(n,k)} = a_{|k|_{\mathbb{Z}_n}}^{\gcd(n,k)} \text{ to the cycle index polynomial of } G \curvearrowright [n].$$

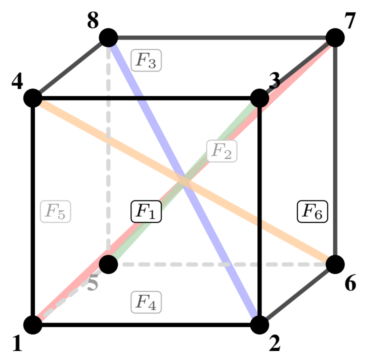
Well, by Lagrange the only possible orders in $G \cong \mathbb{Z}_n$ are divisors d of n, in terms of $d = |k|_{\mathbb{Z}_n} = \frac{n}{\gcd(n,k)}$, we will have that $gcd(n,k) = \frac{n}{d} \implies a_{|k|_{\mathbb{Z}_n}}^{\gcd(n,k)} = a_d^{\frac{n}{d}}$. Finally, since G is cyclic of order n, there are exactly $\varphi(d)$ elements of order d for each divisor d of n. So we can just sum over divisors of n to obtain:

$$Z(G; a_1, \ldots, a_n) = \frac{1}{|G|}\sum_{g\in G}\prod_{i=1}^{n} a_i^{c_i(g)} = Z(\mathbb{Z}_n; a_1, \ldots, a_n) = \frac{1}{n}\sum_{k\in\mathbb{Z}_n} a_{|k|_{\mathbb{Z}_n}}^{\gcd(k,n)} = \frac{1}{n}\sum_{d|n}\varphi(d)a_d^{\frac{n}{d}}.$$

□

Problem 11(a). Compute the cycle index polynomial for the rotational symmetries of the cube acting on faces.

Proof. Let X be the set of faces of a cube, so $X \longleftrightarrow [6]$, and let G be the group of rotational symmetries of the cube, which we established is S_4 in the previous problem. Once more we can just define a correspondence between that action on the poles, vertices, and then faces to compute everything.



Diagonal Poles:

$\mathbf{P_1} = \{1,7\}$

$\mathbf{P_2} = \{2,8\}$

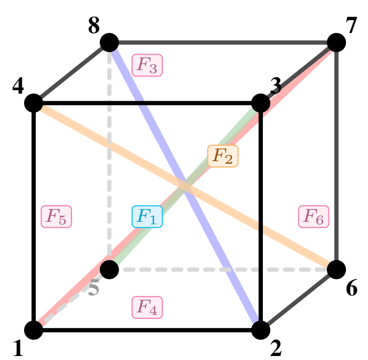
$\mathbf{P_3} = \{3,5\}$

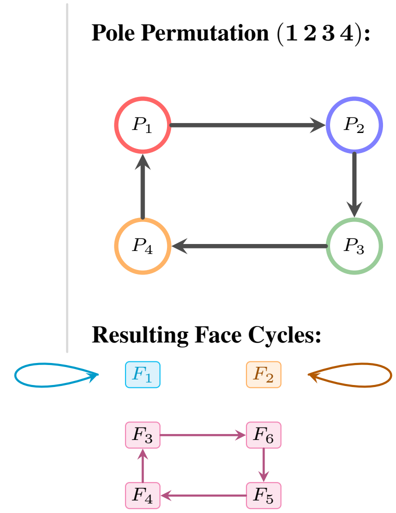
$\mathbf{P_4} = \{4,6\}$

Faces:

Front: $F_1 = (1,2,3,4)$
Back: $F_2 = (5,6,7,8)$
Top: $F_3 = (3,4,8,7)$
Bottom: $F_4 = (1,2,6,5)$
Left: $F_5 = (1,4,8,5)$
Right: $F_6 = (2,3,7,6)$

We show an example below of this correspondence with face permutations





Recall that the monomial $a_1^{c_1(g)}a_2^{c_2(g)}a_3^{c_3(g)}a_4^{c_4(g)}a_5^{c_5(g)}a_6^{c_6(g)}$ for each $g \in G$ count the number of permutations $g \in G$ that result in c_i-many $i-$face cycles for each $i \in [4]$. So the cycle index polynomial Z_G in general counts the number of permutations $g \in G$ that result in each possible face cycle partition/configuration of X. Here we see that (1234) contributes the monomial $a_1^2 a_4$.

Rotation classes and face cycle monomials

Rotation Types	**Count**	**Cycle monomial on faces**
Identity	1	a_1^6
$90°/270°$ about face-center axis	6	$a_1^2 a_4$
$180°$ about face-center axis	3	$a_1^2 a_2^2$
$120°/240°$ about body diagonal	8	a_3^2
$180°$ about midpoint of opposite edges	6	a_2^3

Cycle index polynomial

Summing the right side of the table we get that the cycle index polynomial Z_G of G acting on the 6 faces is

$$Z_G(a_1, a_2, a_3, a_4, a_5, a_6) = \frac{1}{|G|} \sum_{g \in G} a_1^{c_1(g)} a_2^{c_2(g)} a_3^{c_3(g)} a_4^{c_4(g)} a_5^{c_5(g)} a_6^{c_6(g)}.$$

Since $G \cong S_4$, we get that

$$Z_G(a_1, a_2, a_3, a_4, a_5, a_6) = \frac{1}{24}\Big(a_1^6 + 6a_1^2 a_4 + 3a_1^2 a_2^2 + 8a_3^2 + 6a_2^3\Big).$$

This is the desired cycle index polynomial for the action of cube rotations on its faces.

$\square$

Problem 15(a). Consider 4-bead n-colored necklaces under rotation. Find the number of distinct necklaces which have 2 beads of one color and 2 beads of another color in two ways: by using Theorem 6.4.2 and by making a direct count.

Proof. The action here is $\mathbb{Z}_4 \curvearrowright X \longleftrightarrow [4]$. We proved earlier that:

$$Z(\mathbb{Z}_4; a_1, a_2, a_3, a_4) = \frac{1}{4}\sum_{d|4} \varphi(d) a_d^{\frac{4}{d}} = \frac{1}{4}\sum_{d\in\{1,2,4\}} \varphi(d) a_d^{\frac{4}{d}} = \frac{1}{4}(a_1^4 + a_2^2 + 2a_4)$$

Recall:

Theorem 1(a) (Redfield–Pólya Theorem)**.** *Let G be a finite group acting on a finite set X with $|X| = n$, and let Y be a set of variables. Then*

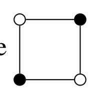
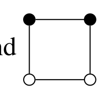
$$\sum_{\mathcal{O}} \mathrm{wt}(\mathcal{O}) = Z\left(G; \sum_{y\in Y} y, \sum_{y\in Y} y^2, \dots, \sum_{y\in Y} y^n\right),$$

where the sum on the left is over the orbits $\mathcal{O}$ of the induced action of G on $Y^X := \{f : Y \to X\}$.

To translate to colorings, basically for a $|Y| = n$-coloring of X, $[y_1^{c_1} \cdots y_n^{c_n}]$ counts the number of elements with c_i of color $i, \forall i \in [n]$. This works for any countable Y too.

For our problem, $Y = \{y_1, y_2\}$ are variables representing our colors, and then we just want to find $[y_1^2 y_2^2]$

$$\begin{aligned}
&\sum_{\mathcal{O} \text{ over } \mathbb{Z}_4 \curvearrowright Y^X} \mathrm{wt}(\mathcal{O}) = Z(\mathbb{Z}_4; (y_1+y_2), (y_1^2+y_2^2), (y_1^3+y_2^3), (y_1^4+y_2^4)) \\
&= \frac{1}{4}((y_1+y_2)^4 + (y_1^2+y_2^2)^2 + 2(y_1^4+y_2^4)) \\
&= \cdots = y_1^4 + y_1^3 y_2 + 2y_1^2 y_2^2 + y_1 y_2^3 + y_2^4 \\
\implies &[y_1 y_2](y_1^4 + y_1^3 y_2 + 2y_1^2 y_2^2 + y_1 y_2^3 + y_2^4) = 2
\end{aligned}$$

Also by direct counting the two bracelets are  and  up to rotations and color choice.

$\square$

Problem 15(b). Consider the cube under rotation where the faces have been colored black and white. Find the generating function for the number of orbits by the number of white and number of black faces in two ways: by using Theorem 6.4.2 and by making a direct count.

Proof. Earlier we found the cycle index polynomial for this action on faces of a cube to be

$$Z(S_4; a_1, a_2, a_3, a_4, a_5, a_6) = \frac{1}{24}\Big(a_1^6 + 6a_1^2a_4 + 3a_1^2a_2^2 + 8a_3^2 + 6a_2^3\Big).$$

So we just use $y = \{y_1, y_2\}$ to plug in $a_i = (y_1^i + y_2^i), \forall i \in [6]$ to obtain the following after simplification

$$\begin{aligned}
&Z(S_4; y_1 + y_2, y_1^2 + y_2^2, y_1^3 + y_2^3, y_1^4 + y_2^4, y_1^5 + y_2^5, y_1^6 + y_2^6) \\
&= \frac{1}{24}\Big((y_1 + y_2)^6 + 6(y_1 + y_2)^2(y_1^4 + y_2^4) + 3(y_1 + y_2)^2(y_1^2 + y_2^2)^2 \\
&\qquad\qquad + 8(y_1^3 + y_2^3)^2 + 6(y_1^2 + y_2^2)^3\Big) \\
&= y_1^6 + y_1^5y_2 + 2y_1^4y_2^2 + 3y_1^3y_2^3 + 2y_1^2y_2^4 + y_1y_2^5 + y_2^6.
\end{aligned}$$

the generating function we want where $[y_1^iy_2^j]$ counts the number of orbits with i faces color 1 and j faces color 2.

To begin counting directly, clearly there is only one way to color all faces black or white, one way to color all faces but one black or white up to rotations. Then for coloring two faces with one color up to rotation, either both faces of the same color are adjacent or opposite, so two ways. Lastly for three faces, we can color three faces in a row, or three faces adjacent forming a corner. These arent distinct orbits for each color, since two monochromatic rows are formed, or two monochromatic corners are formed. Once again, we obtain:

$$y_1^6 + y_1^5y_2 + 2y_1^4y_2^2 + 3y_1^3y_2^3 + 2y_1^2y_2^4 + y_1y_2^5 + y_2^6.$$

□

Problem 31(a). Prove that $e^{\frac{2k\pi i}{n}}$ is a primitive nth root of unity if and only if $\gcd(k, n) = 1$.

Proof. ($\Longrightarrow$) If $e^{\frac{2k\pi i}{n}}$ is a primitive nth root of unity, then $(e^{\frac{2k\pi i}{n}})^m \neq 1, \quad$ for all $0 < m < n$. Now suppose $\gcd(k, n) = d > 1$. Then, $\mathbb{Z}^+ \ni \frac{n}{d} < n$ and $(e^{\frac{2k\pi i}{n}})^{\frac{n}{d}} = (e^{2k\pi i})^{\frac{1}{d}} = (e^{2\pi i})^{\frac{k}{d}} = 1$ since $\frac{k}{d} \in \mathbb{Z}$, a contradiction. So $\gcd(k, n) = 1$.

($\Longleftarrow$) If $\gcd(k, n) = 1$, then $|k|_{\mathbb{Z}_n} = n$. Clearly $e^{\frac{2\pi i}{n}}$ is a primitive nth root of unity, so $\langle e^{\frac{2\pi i}{n}} \rangle \cong \mathbb{Z}_n \implies 1 \longleftrightarrow e^{\frac{2\pi i}{n}}$ is an isomorphism and then $|k|_{\mathbb{Z}_n} = |(e^{\frac{2\pi i}{n}})^k| = n$ as a result and so by definition $e^{\frac{2\pi i}{n}}$ must be a a primitive nth root of unity.

$\square$

Problem 31(b). Prove that the CSP is exhibited by the triple

$$\left(\binom{[n]}{k}, \langle(1\,2\,\cdots\,n)\rangle, \begin{bmatrix} n \\ k \end{bmatrix}_q\right).$$

Proof. Let $X = \binom{[n]}{k}, c = (1\,2\,\cdots\,n), \omega = e^{2\pi i/n}$. We show that for each $d \geq 0$,

$$\left|\{S \in X : c^d(S) = S\}\right| = \begin{bmatrix} n \\ k \end{bmatrix}_q \Big|_{q=\omega^d}.$$

Fix some $d \geq 0$. Earlier we proved that c^d decomposes into $m = \gcd(n,d)$ cycles, each of length $\ell = \frac{n}{m}$.

Suppose that then some cycle $(b_1 \cdots b_\ell)$ in the decomposition of c^d is only partially contained in S and without loss of generality there exists a first consecutive pair (b_i, b_j) such that $b_i \in S, b_j \notin S$. So then b_i is sent outside of S which then can't be fixed by c^d. On the other hand if it is a union of cycles in c^d then of course everything is fixed, because it only travels within a block of S.

So any subset $S \in \binom{[n]}{k}$ is fixed by c^d if and only if it is some union of cycles (orbits) in the decomposition of c^d. That is if some subset of the cycles in c^d viewed as a collection of blocks partitions S. Therefore, if $\ell \nmid k$, then no such fixed subset exists, because it couldn't be partitioned into blocks of size ℓ, the length of cycles in the decomposition of c^d. If $\ell \mid k$, then $k = \ell r$ for some $r \in \mathbb{Z}^+$ and all c^d-fixable S are obtained by distinctly choosing $r = \frac{k}{\ell}$ of the m cycles. Therefore,

$$\left|\{S \in X : c^d(S) = S\}\right| = \begin{cases} \binom{m}{\frac{k}{\ell}}, & \ell \mid k, \\ 0, & \ell \nmid k. \end{cases}$$

Now ω^d has order $\ell = \frac{n}{\gcd(n,d)}$. So $(\omega^d)^j = 1$ if and only if $\ell \mid j$. Observe.

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \Big|_{q=\omega^d} = \frac{(1-(\omega^d)^n)(1-(\omega^d)^{n-1})\cdots(1-(\omega^d)^{n-k+1})}{(1-\omega^d)(1-(\omega^d)^2)\cdots(1-(\omega^d)^k)}.$$

So if $\ell \nmid k$, then among $n, n-1, \ldots, n-k+1$ there are $\left\lceil \frac{k}{\ell} \right\rceil = \left\lfloor \frac{k}{\ell} \right\rfloor + 1$ multiples of ℓ, since n itself is divisible by ℓ. But among $1, 2, \ldots, k$ there are only $\left\lfloor \frac{k}{\ell} \right\rfloor$ multiples of ℓ since it is not. So the numerator contains exactly one more vanishing factor than the denominator and $\begin{bmatrix} n \\ k \end{bmatrix}_q \Big|_{q=\omega^d} = 0$.

If $\ell \mid k$, we have that $n = \ell a, k = \ell b$ for some $a, b \in \mathbb{Z}^+$ and then

$$\begin{aligned} \begin{bmatrix} n \\ k \end{bmatrix}_q \Big|_{q=\omega^d} &= \frac{(1-q^{\ell a})(1-q^{\ell(a-1)})\cdots(1-q^{\ell(a-b+1)})}{(1-q^\ell)(1-q^{2\ell})\cdots(1-q^{b\ell})}\Bigg|_{q=\omega^d} = \frac{a(a-1)\cdots(a-b+1)}{1\cdot 2\cdots b} = \binom{a}{b} \\ &= \binom{n/\ell = gcd(n,d) = m}{k/\ell}. \end{aligned}$$

Therefore $\left|\{S \in X : c^d(S) = S\}\right| = \begin{bmatrix} n \\ k \end{bmatrix}_q \Big|_{q=\omega^d}$ for all $d \geq 0$ and $(\binom{[n]}{k}, \langle(1\,2\,\cdots\,n)\rangle, \begin{bmatrix} n \\ k \end{bmatrix}_q)$ exhibits the cyclic sieving phenomenon.

$\square$